Giorgio Maria Maioli

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Education

Eindhoven University of Technology — **Eindhoven, NL**
MSc Hons in Data Science & Artificial Intelligence — *Sept 2025 – Exp July 2027*
- **Coursework:** Generative AI Models, Deep Learning, Machine Learning Engineering, Big Data Management

Università degli Studi Roma Tre — **Rome, IT**
BS in Economics and Big Data (Final grade: 109/110) — *Oct 2022 – March 2025*
- GPA: 3.79, **graduated 6 months ahead** of schedule · Minor in Data Science
- **Erasmus+** Merit Scholarship: Mykolas Romeris University, Vilnius (Aug 2023 – Jan 2024)

Experience

Data Scientist Intern — **Amsterdam, NL**
QuantumBlack, AI by McKinsey & Company — *Nov 2026 - Feb 2027*
- Joining **interdisciplinary client teams** of Data Scientists, Data Engineers and ML Engineers to develop **analytics solutions** for clients' business challenges.
- Scope: exploratory data analysis, building and validating **ML models**, implementing **LLM use cases** and **RAG systems**, and supporting client **GenAI adoption**.

AI Engineer Intern — **Rome, IT**
BIP – Business Integration Partners — *Apr 2025 – Jul 2025*
- **Architected and developed a full-stack**, **multi-agent platform** for dynamic document generation, **autonomously managing** the project from concept to a fully functional Proof of Concept within 3 months.
- Engineered a **novel two-stage generation pipeline**: first synthesizing source PDFs into a structured "Knowledge Exoskeleton" via **multimodal LLMs**, using it as high-quality context for document creation.
- Designed and implemented a **modular backend architecture** featuring an AgentFactory for **dynamic prompt engineering** and a ProjectManager for robust **data persistence and management**.

Research & Projects

Honors Research Project: Language-Native World Action Models — *Aug 2026 – Present*
Eindhoven University of Technology
- Developing a language-native extension of **DreamZero** to **jointly predict** video, actions, and text, enabling the model to verbalize its imagined future consistently with its visual and action predictions.
- Designing a staged evaluation on **LIBERO** using **RLinf**: frozen-backbone latent probing, joint SFT with an auxiliary text-generation objective, and RL post-training (PPO/GRPO) to test cross-modal consistency as a failure predictor.

MeditaSplit — bunq Hackathon *Agentic Bill-Splitting App* — GitHub
- Built an end-to-end agentic bill-splitting app in under 24h: photo, **Claude Vision** receipt parsing, real Bunq payment requests, with voice input and multi-turn conversational edits.

Subgroup Discovery with LLMs for Sentiment Analysis — GitHub
- Developed a framework using LLMs to auto-generate features for unstructured data identifying subgroups where sentiment classification models systematically fail.

ML Engineering: Fine-Tuning CNNs, ViTs and LLMs for Medical AI
- **Fine-tuned ResNet-18** (87.1%) and **ViT-B/16** on DermaMNIST; **interpreted predictions** via Grad-CAM and Attention Rollout, exposing clinically relevant failure modes on the Melanoma/Nevi pair.
- Applied **LoRA** (via Optuna HPO) to fine-tune Gemma-3-270M-IT on MedMCQA (+8.6pp over random baseline); built a **LlamaIndex RAG pipeline** that eliminated hallucinations on medical queries.

BSc Thesis: Macrofinancial Effects of US Political Cycles via Deep Learning
- Developed and compared **TFT** and **LSTM** models on a 40-year daily dataset of economic, financial, and sentiment indicators, achieving ∼**6**× better performance with TFT over the LSTM baseline.
- Interpreted results via TFT native attention weights and **LIME**, surfacing regime-dependent political-cycle patterns.

Technical Skills

Programming: Python, R, SQL, MATLAB · **Languages:** Italian (Native), English (C1)
AI & ML Frameworks: PyTorch/lightning, scikit-learn, TensorFlow, Keras, Optuna, Hugging Face, LangChain/Graph
Data Science: Pandas, NumPy, SciPy, Matplotlib, LIME, SHAP, LoRA
Database & Data Management: MySQL & Oracle SQL, ChromaDB, Spark, Hadoop